Machine Learning - Graduate Level Assessment 23 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Statement 1| The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2| The Penn Treebank has been used for language modeling.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
2. Adding more basis functions in a linear model, pick the most probably option:
   (a) Decreases model bias
   (b) Decreases estimation bias
   (c) Decreases variance
   (d) Doesn't affect bias and variance
3. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
4. Neural networks:
   (a) Optimize a convex objective function
   (b) Can only be trained with stochastic gradient descent
   (c) Can use a mix of different activation functions
   (d) None of the above
5. The model obtained by applying linear regression on the identified subset of features may differ from the model obtained at the end of the process of identifying the subset during
   (a) Best-subset selection
   (b) Forward stepwise selection
   (c) Forward stage wise selection
   (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: Linear Logistic Regression requires training data to be linearly separable to produce accurate results.
7. True or False: Pruning a Decision Tree primarily aims to avoid overfitting the training set by simplifying the model.
8. True or False: In PCA, transforming the data to have a zero mean is essential to achieve the same projection as using SVD.
9. True or False: L 2 regularization of linear models tends to create sparser solutions compared to L 1 regularization.
10. True or False: K-fold cross-validation is quadratic in K, indicating that the computation time increases with the square of the number of folds.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the fundamental differences between Bayesian and frequentist approaches in machine learning, particularly focusing on the role and implications of prior distributions in probabilistic models.
12. Discuss the necessary probability distributions required to calculate P(H|E, F) without any conditional independence assumptions. Explain the significance of each distribution in your analysis.

End of Paper